

in the EBRT group. Relative biological effectiveness (RBE) value (CLDR/EBRT) was 1.50. Then we studied the data at 48h after different irradiation of 4Gy. The cell growth inhibition rate, cell death rate, apoptosis rate and percentages of cells in G2/M phase in CLDR group were all higher than those in EBRT group. In the CLDR group, p-H2AX (Ser139) expression was significantly elevated. DNA-PKcs protein level and the phosphorylation in T2609 and S2056 sites were also significantly increased. Meanwhile, we found ku80 protein level decreased and ku70 protein level was not obviously changed. Bcl-2 mRNA level was not changed, while protein level was significantly reduced. Bax mRNA and protein level were significantly increased. Cyclin B1 protein level was reduced.

Conclusions: Compared with EBRT, CLDR was more effective in growth inhibition in A549 lung cancer cells. It downregulated Bcl-2 protein and upregulated expression of Bax mRNA to induce cell apoptosis. Downregulated CyclinB1 leads to induce G2/M phase arrest. But CLDR didn't inhibit the function of DNA-PK complex.

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Episcleral Plaque ¹²⁵I Brachytherapy for Choroidal Melanoma: A Single Institution Retrospective Review

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Purpose: Episcleral plaque brachytherapy represents a standard treatment for appropriate choroidal melanoma patients. Variations in prescription dose and dose-rate exist for reasons of physician preference and logistics.

Materials and Methods: A retrospective chart review of all patients treated with ¹²⁵I plaque brachytherapy for ocular melanoma at the University of Arizona was performed. A total of 140 patient records were the basis for this analysis. Data was obtained on dose and dose-rate to the tumor and sensitive structures. Patients were followed by the treating ocular oncologist for local tumor control, distant metastases, and preservation of vision.

Results: The current series represents a similar patient population to those reported by COMS. Prescription dose varied compared to COMS with tumors less than 5mm thick prescribed to the tumor apex rather than to 5mm. The change in prescription point allowed for reduced irradiation of sensitive structures without loss of tumor control. Additionally, the current series varies from ABS recommendations in regards to dose-rate. Current guidelines recommend the dose rate at the prescription point to range between 60 cGy/hr and 105 cGy/hr. This series included patients with a dose-rate as low as 50.6 cGy/hr. No correlation was found between lower dose-rate and tumor control.

Conclusions: The results of this retrospective single institution review confirm what has been known for 3 decades from the COMS study that ¹²⁵I episcleral plaque therapy is an effective, low morbidity, treatment for medium size and small but rapidly growing choroidal melanomas. For tumors with a thickness less than 5mm, reducing the prescription depth to the tumor apex served to decrease the dose to all sensitive structures within the eye. This dose reduction occurred without any significant loss of local control. Although our dose-rate varied somewhat from the ABS guidelines no differences in either the control or the complications rates were noted as a function of dose-rate.

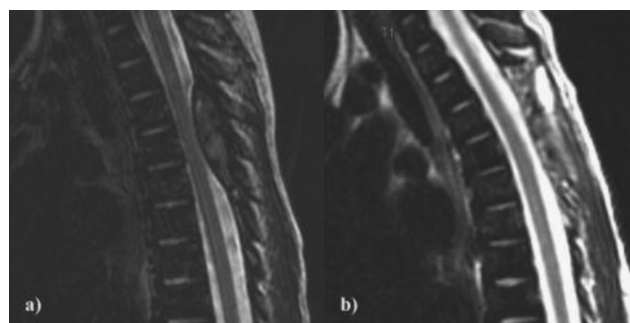
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Case Report: 6-Year Old Girl with Multiply Recurrent Neuroblastoma Treated with Intraoperative ³²P Brachytherapy Plaque

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Purpose: Achieving local control for neuroblastoma is a crucial component in the management of the disease, but this may be complicated in the setting of prior radiation treatment, especially when the therapeutic target is in the proximity of critical structures such as the spinal cord. We describe a pediatric patient with multiply recurrent neuroblastoma and prior high-dose radiation therapy to the spine who presented with progressive epidural disease. She was managed with surgical resection and high-dose-rate (HDR) intraoperative brachytherapy using a ³²P plaque previously developed for treatment of brain and spine lesions.

Materials and Methods: A 6-year old girl with multiply recurrent stage IV neuroblastoma treated with numerous lines of chemotherapy was referred to our radiation oncology and neurosurgical services due to progressive disease in her thoracic spine, causing spinal cord compression. She had been treated with hypofractionated image-guided external beam radiation therapy to the spine at the site of progressive disease 11 months prior, receiving a dose of 25 Gy in 5 fractions. On presurgical planning MRI, an epidural mass was visualized from T3 to T7, compressing and displacing the spinal cord. Surgical resection was recommended, but due to the high likelihood of microscopic residual disease at the dural margin and concern for toxicity resulting from repeated external beam radiation to the cord, the multidisciplinary tumor board recommended treatment of the involved dura with intraoperative brachytherapy. Use of the ³²P intraoperative brachytherapy plaque was recommended due to the rapid dose falloff with depth, with a percent depth-dose of <5% of prescription dose after 3 mm of cerebrospinal



fluid (CSF) penetration. Prior to treatment, the thickness of the CSF within the thecal sac was measured proximal and distal to the lesion to ensure adequate attenuation of electrons to minimize the dose to the spinal cord.

Results: Measurement of the CSF thickness from the interior surface of the dura to the surface of the cord proximal and distal to the site of compression demonstrated >3mm thickness (4.7mm and 4.5mm, respectively). Dural thickness was approximately 1mm. Thoracic laminectomy of the T3-T6 vertebral levels with excision of extradural tumor was performed to prevent impending neurological damage from cord compression and achieve local control. The involved dura was treated with intraoperative brachytherapy using a 1 x 6 cm ³²P plaque applied along the posterior aspect of the surgically exposed dura. A dose of 10 Gy was prescribed to 1 mm depth at a dose rate of 71.6 cGy/minute over a total of 13 minutes and 58 seconds. Postoperative MRI showed gross total resection of the lesion with expected postoperative changes and decompression of the underlying spinal cord.

There were no acute postoperative complications secondary to the surgical procedure or intraoperative radiation therapy.

Conclusions: Intraoperative brachytherapy with a ³²P plaque was successfully used in the surgical management of a previously irradiated recurrent thoracic lesion in a 6 year old girl with stage IV neuroblastoma. A dose of 10 Gy was delivered to the dural margin with <1% of prescription delivered to the surface of the previously treated spinal cord. This represents the first use of the ³²P plaque in this age group.